Derivation of $\frac{dL}{dz} = a - y$ (Optional Calculus)

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Goal and Setup

- We aim to derive $\dfrac{dL}{dz} = a - y$ for logistic regression.
- Definitions for a single training example:

$$a = \sigma(z) = \frac{1}{1 + e^{-z}},$$
$$L(a, y) = -\big[y \log a + (1 - y) \log(1 - a)\big].$$

Step 1: Derivative of L with respect to a

Starting from $L(a, y) = -\big[y \log a + (1 - y) \log(1 - a)\big]$:

$$\begin{aligned} \frac{dL}{da} &= -\frac{y}{a} + \frac{1 - y}{1 - a} \\ &= \frac{a - y}{a(1 - a)}. \end{aligned} \tag{$*$}$$

(Used: $\frac{d}{da} \log a = \frac{1}{a}$ and $\frac{d}{da} \log(1 - a) = -\frac{1}{1-a}$.)

Step 2: Derivative of a with respect to z

Using the sigmoid derivative:

$$\frac{da}{dz} = \frac{d}{dz}\sigma(z) = \sigma(z)\big(1 - \sigma(z)\big) = a(1 - a). \tag{$\dagger$}$$

Step 3: Chain Rule to get $\frac{dL}{dz}$

Apply the chain rule:

$$\begin{aligned}\frac{dL}{dz} &= \frac{dL}{da} \cdot \frac{da}{dz} \\ &\overset{(*,\dagger)}{=} \frac{a-y}{a(1-a)} \cdot a(1-a) \\ &= a - y.\end{aligned}$$

Result: $\boxed{\frac{dL}{dz} = a - y}$.

Meaning of $a - y$

- a is the predicted probability; $y \in \{0, 1\}$ is the true label.
- $a - y$ is the **error term** at the logit z.
- It drives gradients for parameters:

$$\frac{\partial L}{\partial W} = X\,(a - y), \qquad \frac{\partial L}{\partial b} = a - y.$$

- Gradient descent updates: $W \leftarrow W - \alpha\, X(a - y)$, $b \leftarrow b - \alpha\,(a - y)$.

Step	Expression	Outcome
1	$\frac{dL}{da} = -\frac{y}{a} + \frac{1-y}{1-a}$	$\frac{a-y}{a(1-a)}$
2	$\frac{da}{dz} = a(1-a)$	Sigmoid slope
3	$\frac{dL}{dz} = \frac{dL}{da} \cdot \frac{da}{dz}$	$\boxed{a-y}$